NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

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Project Report

Sindh Multi-City Climate Prediction System Using Deep Learning (LSTM)

Course: Deep Learning

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ABSTRACT

This Project presents a Multi-City Climate Prediction System that uses deep learning techniques to estimate temperature patterns in 15 cities in Sindh, Pakistan. In order to capture temporal dependencies in historical climate data from 2000 to 2024, the system is built on Long Short-Term Memory (LSTM) networks. In order to improve generality and lessen model-specific bias, consistent meteorological variables were extracted from several high-resolution climate model outputs, such as CMCC_CM2_VHR4 and six other CMIP-based climate simulation models.

A high-resolution climate model (CMCC_CM2_VHR4) was used to create the dataset, from which important meteorological parameters like temperature, humidity, wind speed, and precipitation were taken. In order to replicate regional climatic variability across cities like Karachi, Hyderabad, Sukkur, Jacobabad, and Tharparkar, a multi-city framework was developed by adding temperature biases and stochastic noise levels that were prompted by science.

Missing values were handled, invalid timestamps were eliminated, MinMaxScaler was used for normalization, and a sliding window technique (lookback 7–14 days depending on city) was used to transform the data into supervised learning sequences. In order to improve realism, 5% fictitious missing values (-999) were introduced and then imputed using the mean temperature for each city.

With city-specific hyperparameters, such as different LSTM units, lookback times, and training epochs, distinct LSTM models were developed for every city. The R2 score and Root Mean Squared Error (RMSE) were used to assess the model's performance.

The findings show that predictability varies significantly amongst cities, with stable climates outperforming extremely volatile areas like Jacobabad in terms of accuracy. The impact of climate variability on prediction performance was further examined through cross-city comparison and seasonal RMSE evaluation.

INTRODUCTION

Problem Context:

Accurate temperature forecasting is becoming more crucial for urban planning, agriculture, and disaster management due to climate change and regional weather variability. Due to desert, maritime, and interior influences, climate conditions in places like Sindh, Pakistan, differ greatly across short geographic distances.

Nonlinear temporal connections in climate data are difficult for traditional statistical forecasting methods to grasp. As a result, deep learning models are frequently used for time-series forecasting problems, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks.

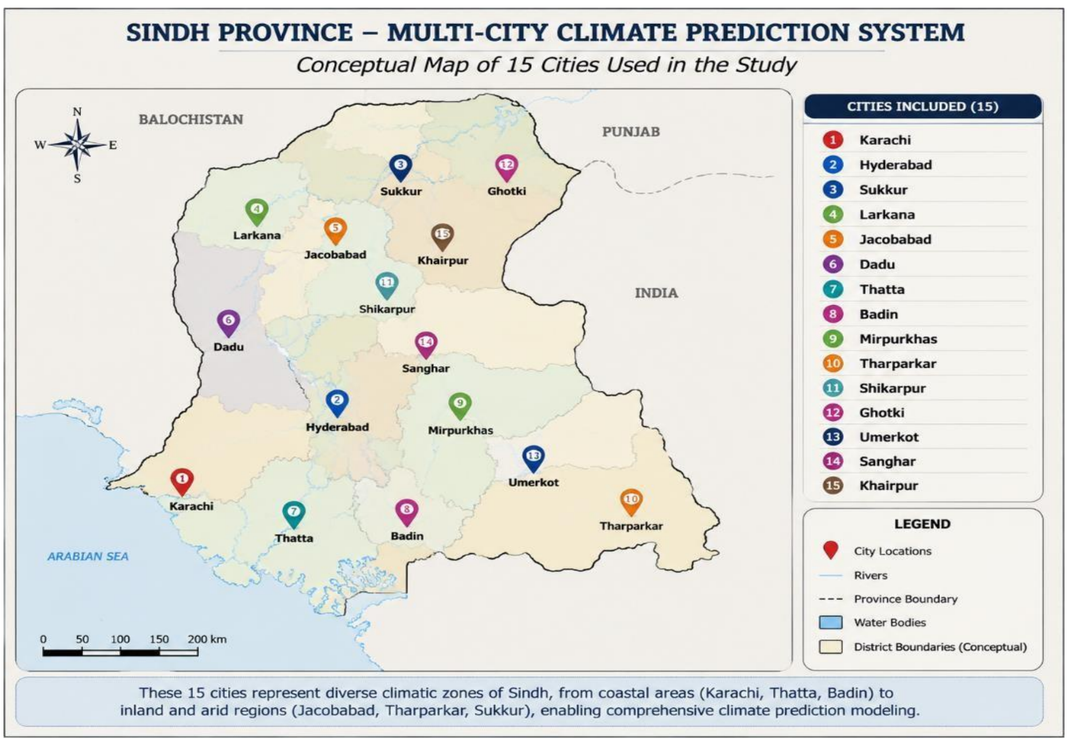

Fig 1: 15 Sindh cities included in the multi-city climate prediction system.

Motivation:

The goal of this project is to develop a multi-city forecasting system that is scalable & capable of:

- Acquires knowledge of temporal weather dependencies
- adjusts to variations in the local climate
- offers predictive behavior specific to a city.
- assesses regional variations in climate predictability.

Literature Insights:

Due to its significance in agriculture, disaster management, and environmental planning, timeseries forecasting of climatic variables has been an active research subject. Early climate prediction systems made extensive use of conventional statistical models like ARIMA and exponential smoothing, but these models have trouble capturing nonlinear and long-term dependencies found in meteorological data. Recurrent Neural Networks (RNNs) are now a common option for sequential data modeling due to the development of deep learning. Nevertheless, the vanishing gradient problem hinders typical RNNs' capacity to discover long-term correlations in lengthy climatic sequences. Hochreiter and Schmidhuber introduced Long Short-Term Memory (LSTM) networks in 1997 to get around this restriction by adding memory cells and gating mechanisms that control information flow over lengthy sequences [1]. When it comes to modeling intricate temporal correlations in meteorological information, this design has shown great efficacy. The superiority of LSTM-based models in time-series forecasting tasks has been shown in recent studies. When compared to traditional RNN models, Qin et al.'s attention-based LSTM encoder– decoder framework greatly increased multivariate time-series prediction accuracy [2]. In a similar vein, Siami-Namini et al. discovered that LSTM networks frequently beat ARIMA in nonlinear forecasting tasks when comparing deep learning models with conventional statistical techniques [3]. Because deep learning can capture intricate relationships between atmospheric variables like temperature, humidity, wind speed, and precipitation, it has becoming more and more popular in climate research applications. Research has demonstrated that multivariate deep learning models can enhance prediction accuracy in regional forecasting tasks by efficiently learning these interdependencies. Additionally, to lessen bias and uncertainty in climate projections, ensemble climate datasets from CMIP (Coupled Model Intercomparison Project) simulations have been extensively utilized. Multi-model ensembles are crucial for enhancing the robustness of climate projections, according to the Intergovernmental Panel on Climate Change (IPCC) [4]. Furthermore, as global models frequently fall short of accurately capturing regional weather fluctuations, recent research shows that region-specific or localized training enhances forecasting ability in heterogeneous climate zones.

These results are applied in this investigation by:

- Sequential climate prediction with LSTM networks
- To address spatial heterogeneity, distinct models are trained for every city.
- Reducing bias by utilizing several climate model outputs
- Assessing performance across several climatic zones in Sindh using RMSE and R2 measures

METHODOLOGY

Data Collection and Feature Selection:

The dataset is derived from multiple climate simulation models, including CMCC_CM2_VHR4 and six additional CMIP-based models. This ensemble approach reduces bias and improves robustness in climate representation.

Selected features:

- Temperature (target variable)
- Wind speed
- Relative humidity
- Precipitation

CMCC_CM2_VHR4 was used as the reference model for preprocessing due to its high resolution and stable performance.

Multi-City Data Generation:

A synthetic multi-city dataset was created by:

- Applying city-specific temperature bias
- Adding Gaussian noise based on climate variability
- Modifying humidity and precipitation based on temperature relationships

This produced realistic climate variation across 15 Sindh cities.

Missing Data Handling:

To simulate real-world conditions:

- 5% temperature values were replaced with -999
- Missing values were imputed using city-wise mean temperature

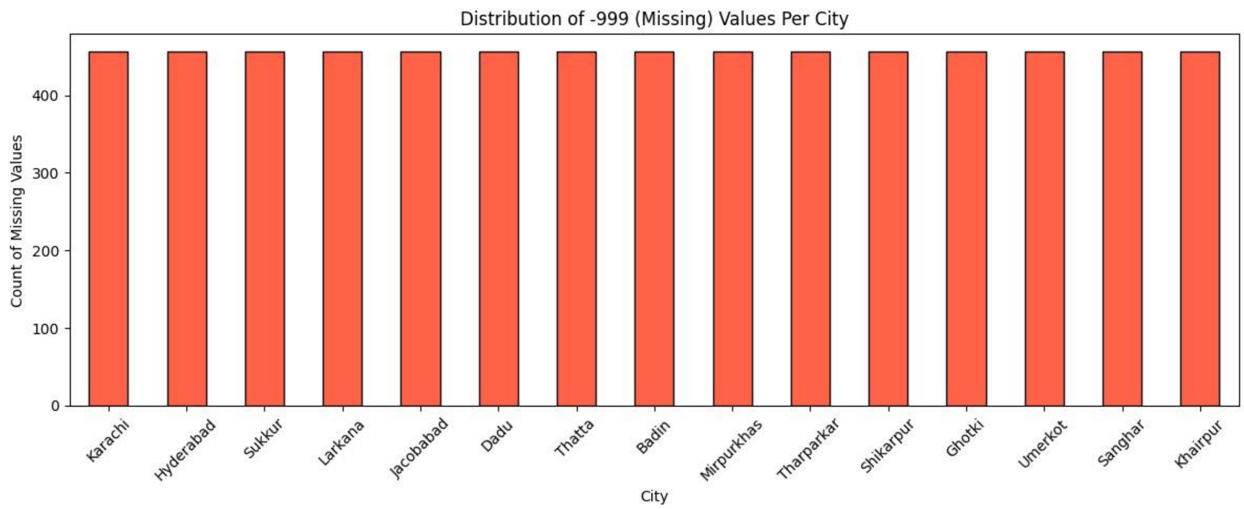

fig 2: Distribution of injected missing values across cities

Feature Scaling:

All numerical features were normalized using MinMaxScaler to improve neural network convergence.

Sequence Generation:

Time-series data was converted into supervised learning format using a sliding window approach:

- Lookback: 7–14 days (city-dependent)
- Input: past observations
- Output: next-step temperature prediction

Model Architecture:

Each city uses a separate LSTM model:

- LSTM Layer 1: 64–128 units
- Dropout: 0.2
- LSTM Layer 2: half units
- Dropout: 0.2
- Dense layer: 1 output neuron

Optimizer: Adam

Loss Function: MSE

City-wise Hyperparameters:

Different cities use different configurations based on climate variability:

- Stable cities → smaller models
- Extreme climates → larger models + longer lookback

City	LSTM Units	Lookback (days)	Epochs	Tier
Jacobabad	128	14	25	High
Tharparkar	128	14	25	High
Sukkur	128	10	20	Mid
Larkana	128	10	20	Mid
Shikarpur	128	10	20	Mid
Khairpur	128	10	20	Mid
Karachi	64	7	15	Low
Hyderabad	64	7	15	Low
Dadu	64	7	15	Low
Thatta	64	7	15	Low
Badin	64	7	15	Low
Mirpurkhas	64	10	15	Low
Ghotki	64	10	15	Low
Umerkot	64	7	15	Low
Sanghar	64	7	15	Low

Table 1: City-wise Hyperparameter Configuration

EXPERIMENTAL SETUP

Train-Test Split:

- Time-series split (no shuffling)
- 80% training, 20% testing

Evaluation Metrics:

The performance of the proposed climate prediction model is evaluated using two standard regression metrics: Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2 Score). These metrics help measure prediction accuracy and model reliability.

1. **Root Mean Squared Error (RMSE)**

RMSE measures the average prediction error between actual and predicted temperature values. It calculates how far the predicted values deviate from the real observation**s.** <u>A lower RMSE value indicates better prediction accuracy and improved model performance.</u>

2. **Coefficient of Determination (R^2 Score)**

The R^2 score measures how well the predicted values fit the actual observations. It represents the proportion of variance in temperature explained by the model. <u>Values closer to **1** indicate stronger predictive performance</u>, while values closer to **0** indicate weaker predictive capability. Negative values suggest poor model performance.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y})^2}{\sum(y_i - \bar{y})^2}$$

Where,

$\hat{y}$ – predicted value of y

$\bar{y}$ – mean value of y

Validation Strategy:

- Separate model per city
- Independent scaling per city
- City-wise evaluation

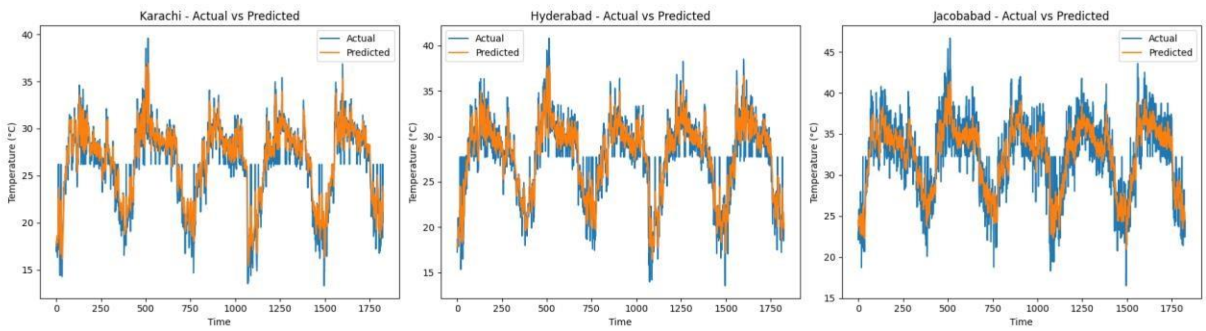

Fig 3: Actual vs Predicted Temperature Comparison

This figure demonstrates that the model generalizes well across multiple cities and is capable of capturing temperature trends with reasonable accuracy, supporting its effectiveness for multi-city climate prediction. The LSTM model shows good performance across all three cities. In Karachi, the predicted temperature closely follows the actual values, effectively capturing seasonal patterns and short-term variations, with only slight deviations during extreme peaks and drops. In Hyderabad, there is a strong alignment between actual and predicted values, and the model remains consistent over most time steps, with only minor lag in highly fluctuating regions. For Jacobabad, the model follows the overall trend well but shows slightly higher deviations compared to the other cities, with occasional underestimation and overestimation during extreme temperature changes.

Visualization Techniques:

- Actual vs Predicted plots
- Correlation heatmap
- City-wise RMSE comparison
- Seasonal error analysis

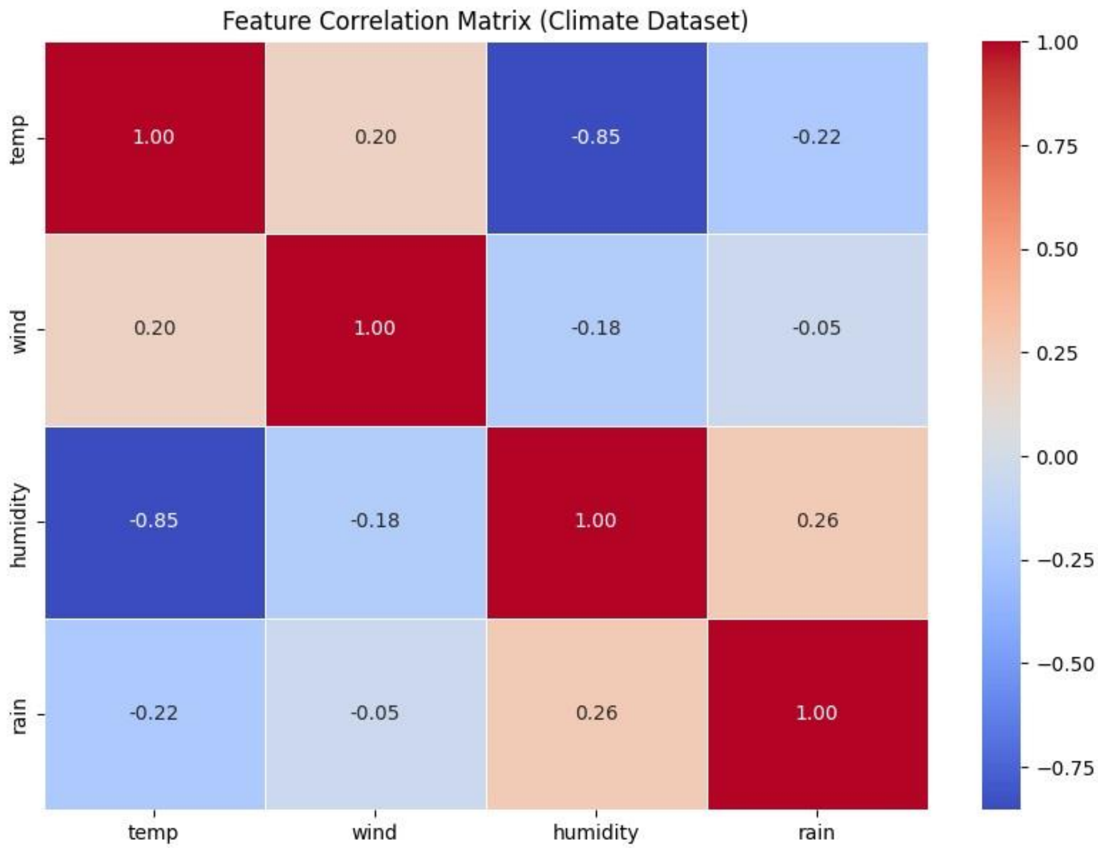

Fig 4: Feature Correlation Matrix

The correlation matrix shows the relationship between climate features in the dataset. Temperature and humidity have a strong negative correlation of −0.85, which means that as temperature increases, humidity decreases. Wind has a weak positive correlation with temperature (0.20) and almost no relationship with rain (−0.05). Humidity and rain show a weak positive correlation (0.26). Overall, the heatmap helps in understanding how different climate variables are connected and which features are more important for prediction.

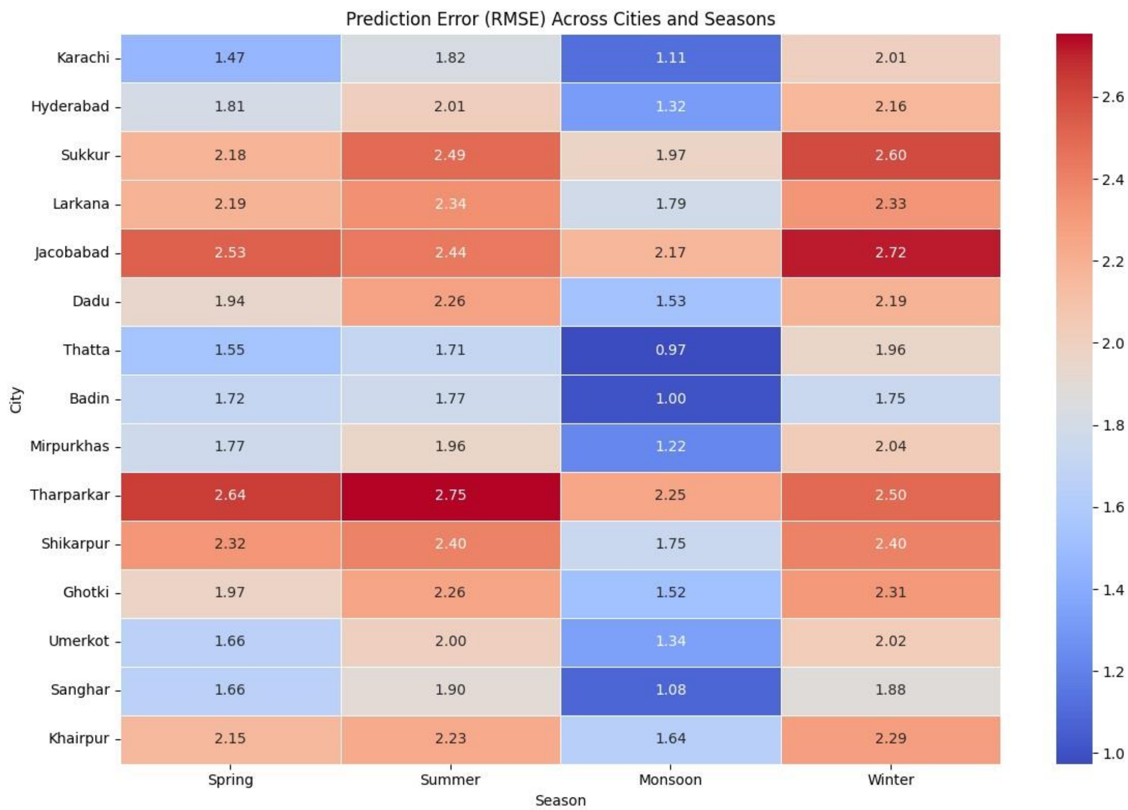

Table 2: Seasonal RMSE across cities.

The heatmap shows the RMSE prediction error of different cities across all seasons. Lower RMSE values indicate better model performance, while higher values represent larger prediction errors. The model performed best during the Monsoon season for most cities, with Thatta showing the lowest RMSE value (0.97). In contrast, cities like Tharparkar and Jacobabad showed higher RMSE values, especially during Summer and Winter, indicating more difficult climate prediction in extreme weather conditions. Overall, the results show that prediction accuracy varies across cities and seasons.

RESULTS

Overall Performance:

Model performance varies significantly across cities due to climatic diversity.

- Stable climate cities → higher R^2, lower RMSE
- Extreme climate cities → lower accuracy

City	RMSE	R^2 Score
Badin	1.646839	0.863894
Thatta	1.696774	0.853730
Sanghar	1.740186	0.850937
Karachi	1.753002	0.845708
Mirpurkhas	1.858982	0.829293
Umerkot	1.861847	0.825521
Hyderabad	1.947070	0.815208
Dadu	2.070563	0.791002
Ghotki	2.126722	0.781100
Khairpur	2.154443	0.774584
Larkana	2.226573	0.773962
Shikarpur	2.290126	0.759636
Sukkur	2.407878	0.738223
Jacobabad	2.531341	0.723638
Tharparkar	2.548250	0.714676

Table 3: Accuracy Report of R2 and RMSE for All 15 Cities:

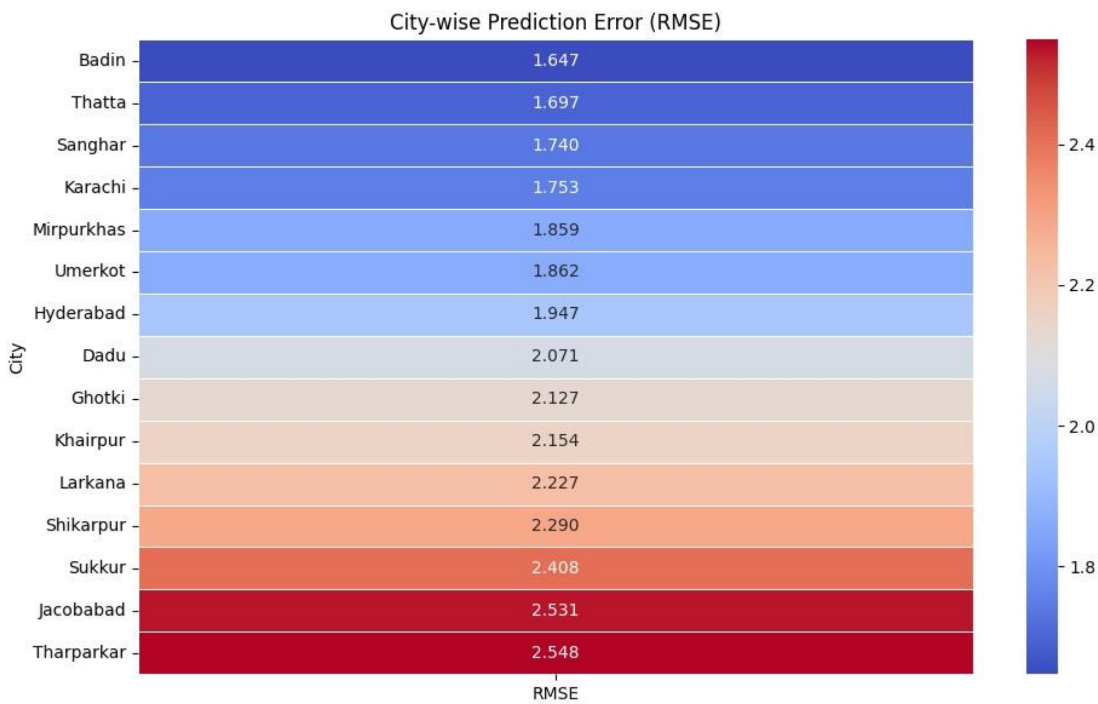

Fig 5: City-wise RMSE analysis

Best and Worst Cities:

- Best predictable city: **Badin**
- Worst predictable city: **Tharparkar**

Seasonal Performance:

- Summer/Monsoon → highest error
- Winter → most stable predictions

Key Findings:

- Temperature variability strongly affects accuracy
- LSTM performs better in stable patterns
- City-specific modeling improves performance

DISCUSSION

Challenges:

- Synthetic dataset introduces approximation bias
- Extreme weather increases prediction error
- Seasonal fluctuations reduce consistency

Limitations:

- No real-world sensor validation
- Only temperature is predicted
- No external meteorological API integration

Future Work:

- Real-world climate API integration
- Transformer-based models
- Multi-variable forecasting system
- Deployment as climate dashboard

CONCLUSION

This project successfully developed a Multi-City Climate Prediction System using LSTM networks. By simulating 15 cities with realistic climate variability and using multiple climate model outputs, the system demonstrates how deep learning can effectively model temporal climate patterns.

Results show that prediction accuracy varies significantly based on climate stability, confirming that regional modeling improves performance compared to a global model. Seasonal analysis further highlights the complexity of climate forecasting in dynamic environments.

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